

Problem #3: Mnemonic Major System (15 marks)

The Mnemonic Major System is a technique used by mental athletes who take part in memory championships. This technique is used to convert numbers to meaningful words that are easy to visualize, and hence easy to memorize. Listed below are a few examples.

314,159,265,358: **meteor tail banshee lime loaf**

75,141,195: **cluttered table**

701,894: **ghost vapour**

99,501,247: **pebble stone rock**

7,512,026: **golden snitch**

8: **hive**

854: **flower**

2,394,429: **number rainbow**

6,096: **cheese bush**

Note: The commas are not related to the encoding. They just make the numbers easy to read.

Convert these phrases into numbers:

- a) **short striped zebra**
- b) **moon cash**
- c) **cat dog**
- d) **grey elephant**

The earliest version of the Mnemonic Major System was designed by Pierre Hérigone (1580–1643), a French mathematician and astronomer. The name "major system" refers to Major Beniowski, who published a version of the system in his book, *"The Anti-Absurd or Phrenotypic English Pronouncing and Orthographical Dictionary"*.

Similar number memorization systems also occur in other parts of the world, such as the *Katapayadi* system going back to at least the 7th Century AD in India.